

2026 GOLD GLOBAL STRATEGY FOR COPD

Diagnosis, Spirometry Criteria & Severity Assessment

1. DIAGNOSTIC CRITERIA & PATHOPHYSIOLOGY

Chronic Obstructive Pulmonary Disease (COPD) is a common, preventable, and treatable respiratory disorder characterized by chronic respiratory symptoms (dyspnea, cough, sputum production) resulting from abnormalities of the airways (bronchitis) and/or alveoli (emphysema).

SPIROMETRIC CONFIRMATION:

A post-bronchodilator FEV1/FVC ratio < 0.70 is mandatory to confirm persistent airflow limitation.

GOLD SPIROMETRIC GRADING (Based on post-BD FEV1 % predicted):

- ? GOLD 1 (Mild): $FEV1 \geq 80\%$ predicted.
- ? GOLD 2 (Moderate): $50\% \leq FEV1 < 80\%$ predicted.
- ? GOLD 3 (Severe): $30\% \leq FEV1 < 50\%$ predicted.
- ? GOLD 4 (Very Severe): $FEV1 < 30\%$ predicted.

2. CLINICAL ASSESSMENT (ABE ASSESSMENT TOOL)

Combines symptom burden (mMRC or CAT score) and exacerbation history:

- ? Group A: 0 to 1 moderate exacerbation not leading to hospital admission, low symptoms (mMRC 0-1, CAT < 10).
- ? Group B: 0 to 1 moderate exacerbation, high symptoms (mMRC ≥ 2 , CAT ≥ 10).
- ? Group E: ≥ 2 moderate exacerbations or ≥ 1 exacerbation leading to hospital admission, regardless of symptom score.

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Pharmacologic Maintenance: Bronchodilators & Inhaled Corticosteroids

3. MAINTENANCE BRONCHODILATION

Long-acting bronchodilators are central to symptom management and exacerbation reduction in COPD.

DUAL BRONCHODILATION (LABA + LAMA):

Dual therapy with a Long-Acting Muscarinic Antagonist (LAMA) plus a Long-Acting Beta2-Agonist (LABA) is superior to monotherapy and is recommended as initial maintenance therapy for Groups B and E.

? Tiotropium (LAMA): 18 mcg inhalation once daily

(HandiHaler) or 5 mcg once daily (Respimat).

? Formoterol (LABA): 12 mcg inhalation twice daily.

? Fixed-dose combinations: Umeclidinium/Vilanterol (62.5/25 mcg once daily) or Tiotropium/Olodaterol (5/5 mcg once daily).

4. INHALED CORTICOSTEROIDS (ICS) GUIDANCE

ICS should NOT be used as monotherapy in COPD. Triple therapy (LABA + LAMA + ICS) is indicated under specific criteria:

? Blood Eosinophils ≥ 300 cells/uL: Strong recommendation for ICS addition.

? Blood Eosinophils 100 to 299 cells/uL WITH ≥ 2 moderate exacerbations per year: Consider ICS addition.

? Blood Eosinophils < 100 cells/uL: ICS is NOT recommended due to lack of efficacy and increased risk of pneumonia.

RECOMMENDED TRIPLE REGIMEN:

Fluticasone Furoate / Umeclidinium / Vilanterol (100/62.5/25 mcg once daily) or Budesonide / Glycopyrrolate / Formoterol (320/18/9.6 mcg twice daily).

TABLE 1: COPD MAINTENANCE REGIMENS

Group A: Single bronchodilator (SABA or LABA)

Group B: LABA + LAMA dual bronchodilation

Group E (Eos < 300): LABA + LAMA dual therapy

Group E (Eos ≥ 300): LABA + LAMA + ICS triple therapy

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Acute Exacerbations: Systemic Steroids, Antibiotics & Oxygen Target

5. ACUTE EXACERBATIONS MANAGEMENT

A COPD exacerbation is defined as acute worsening of dyspnea, cough, and/or sputum within ≤ 14 days.

SYSTEMIC CORTICOSTEROIDS:

? Oral Prednisone 40 mg once daily for 5 days is recommended.

? Prolonged courses (> 5 days) do not improve outcomes and substantially increase adverse effects.

ANTIBIOTIC THERAPY (5 TO 7 DAYS):

Indicated if patient exhibits all three cardinal symptoms (increased dyspnea, increased sputum volume, and increased sputum purulence) or two symptoms if one is purulent sputum:

? First-line: Amoxicillin-Clavulanate 875/125 mg orally twice daily, or Azithromycin 500 mg day 1, then 250 mg daily.

6. CONTROLLED OXYGEN THERAPY

In acute exacerbations, supplemental oxygen should be titrated to achieve a target oxygen saturation (SpO_2) of 88% to 92% in patients at risk of hypercapnic respiratory failure.

CAUTION: Excessive hyperoxia blunts hypoxic drive, worsens V/Q mismatch, and induces severe hypercapnia and acidosis.

7. NON-INVASIVE VENTILATION (NIV)

Bi-level positive airway pressure (BiPAP) is the first-line intervention for acute hypercapnic respiratory failure ($pH < 7.35$ and $PaCO_2 > 45$ mmHg), reducing intubation rates and mortality.